

The Duality of Scale and Space: Wavelet Invariants and Discrete Morse Saddles in Controlled Swarm SDS

System Architecture Invariant Manifest

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1 The Architectural Duality: DWT and DMT

Deterministic closed-loop control over a 100-mile turbulent grid is achieved by separating scale-invariant trends from localized geometric bottlenecks, eliminating all statistical computing.

1.1 The Wavelet Scaling Invariant

The weather advection field is governed by non-linear Navier-Stokes equations, presenting a multi-scale disturbance space U . We apply a Discrete Wavelet Transform (DWT) as a deterministic algebraic filter. The DWT decomposes the field into scale approximations:

$$\psi_{j,k}(x, y) = 2^{-j/2} \psi(2^{-j}x - k_x, 2^{-j}y - k_y) \quad (1)$$

The wavelet transform extracts the macro-system scale trends (low-frequency components) needed to anticipate global battery decay profiles. However, it does not locate the exact coordinates of micro-turbulence cells due to temporal-spatial frequency localization bounds.

1.2 Discrete Morse Coordinate Pinpointing

To map the exact spatial positions of barriers and line-of-sight signal occlusions, a non-degenerate Discrete Morse Function $f : K \rightarrow \mathbb{R}$ is evaluated over the communication complex $\mathcal{G}_C$. While the wavelets provide the scaling envelope, the Morse function uncovers the precise coordinate locations of the critical simplices:

- **Critical Minima Vertices (c_0):** Coordinate cells tracking total network isolation traps.
- **Critical Saddle Edges (c_1):** The exact spatial windows—such as the verified tunnel $(0, 2)$ —where multi-hop communication lines bypass non-convex obstacles.

1.3 Delay-Tolerant Store-and-Forward Buffering

When an asset enters a line-of-sight occlusion zone mapped by the Morse function, a store-and-forward capability caches visual sensor data locally within a computational queue state Q_{local} . The Self-Similar Permutation Group pauses the active routing word transaction until the drone’s trajectory slides along the V-path gradient streamlines and re-enters the coordinates of a valid critical saddle tunnel, flushing the cached data packet with zero loss.